Coffee Shop Sales Analysis & Business Intelligence Dashboard Report

Project Overview

The Coffee Shop Sales Analysis & Business Intelligence Dashboard project was developed to analyze large-scale transactional sales data and transform raw business records into actionable insights. The project focuses on identifying sales trends, customer purchasing behavior, operational performance, and product-level profitability using business intelligence and data analytics techniques.

The primary objective of this project was to support data-driven decision-making by building a professional dashboard capable of monitoring key business metrics and operational performance across multiple store locations.

The project demonstrates practical skills in:

- Data Cleaning
- Data Analysis
- KPI Reporting
- Dashboard Development
- Business Intelligence
- Sales Analytics
- Data Visualization
- Trend Analysis

The dashboard was designed to provide business stakeholders and management teams with an interactive and easy-to-understand overview of company performance.

Business Problem Statement

Coffee shop businesses generate large volumes of transactional data daily. However, raw sales data alone does not provide meaningful insights unless it is properly cleaned, processed, and analyzed.

The business required a centralized reporting solution to:

- Monitor revenue performance
- Identify best-selling products
- Analyze customer purchasing patterns
- Evaluate branch performance
- Detect peak operational hours
- Support strategic business decisions

This project addresses these challenges by transforming raw transactional records into an interactive business intelligence dashboard.

Project Objectives

The main objectives of the project were:

- Analyze overall sales performance
- Track monthly revenue trends
- Identify top-performing products
- Measure store-level performance
- Detect peak sales hours
- Understand customer purchasing behavior
- Build KPI-driven dashboards
- Improve business reporting and visualization
- Support operational and strategic decision-making

Dataset Overview

Dataset Information

Attribute	Description
Dataset Name	Coffee Shop Sales Dataset
Dataset Type	Transactional Sales Data
Total Records	149,000+ Transactions
Data Source	Retail Coffee Shop Sales Records

Main Columns Used

Column Name	Description
transaction_id	Unique transaction identifier
transaction_date	Date of transaction
transaction_time	Time of transaction
transaction_qty	Quantity sold
store_location	Branch/store location
product_category	Product category
product_type	Product type
product_detail	Product description
unit_price	Unit price of product

Column Name	Description
Revenue	Total transaction revenue
Month	Extracted month from date
Hour	Extracted hour from transaction time
Weekday	Extracted weekday from transaction date

Data Cleaning & Preparation

Data cleaning and preprocessing were essential steps in ensuring the quality and reliability of the analysis.

Several data quality issues were identified within the raw dataset, including inconsistent formatting, missing values, and duplicate records.

Data Cleaning Steps Performed

- Checked and handled missing values
- Removed duplicate transaction records
- Verified transaction IDs for consistency
- Standardized date and time formats
- Created calculated revenue column
- Extracted month from transaction dates
- Extracted hourly information from transaction times
- Generated weekday columns for trend analysis
- Validated data consistency and accuracy

Revenue Formula

```
=transaction_qty * unit_price
```

Key Performance Indicators (KPIs)

Several business KPIs were developed to evaluate operational and financial performance.

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1. Total Revenue

Definition

Measures the total sales revenue generated by the business.

Formula

```
=SUM(Revenue)
```

Business Importance

- Tracks overall business growth
- Measures financial performance
- Helps evaluate sales effectiveness

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2. Total Orders

Definition

Measures the total number of customer transactions.

Formula

```
=COUNTA(transaction_id)
```

Business Importance

- Indicates customer traffic
- Measures operational activity
- Supports workforce planning

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3. Average Order Value (AOV)

Definition

Calculates the average spending per customer transaction.

Formula

```
=Total Revenue / Total Orders
```

Business Importance

- Helps understand customer purchasing behavior
- Supports pricing strategy
- Measures customer spending patterns

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4. Best-Selling Product

Definition

Identifies products generating the highest sales revenue.

Business Importance

- Supports inventory management
- Helps optimize product offerings
- Improves menu strategy

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5. Peak Sales Hour

Definition

Identifies the busiest operational hour during the day.

Business Importance

- Improves staffing efficiency
- Helps optimize operational scheduling
- Supports workforce allocation

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6. Best Performing Store

Definition

Measures the highest revenue-generating branch location.

Business Importance

- Evaluates branch performance
- Identifies expansion opportunities
- Supports operational improvement strategies

Dashboard Analysis & Insights

1. Monthly Revenue Analysis

Monthly sales analysis was conducted to identify seasonal patterns and revenue fluctuations.

Key Findings

- Revenue varies significantly across different months

• Certain months generate stronger business performance
• Seasonal trends influence customer purchasing behavior

Business Impact

This analysis helps management forecast future sales and improve inventory planning.

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2. Product Category Analysis

Sales performance was analyzed across multiple product categories.

Product Categories

• Coffee
• Tea
• Bakery
• Chocolate
• Drinking Chocolate

Key Findings

• Coffee products generated the highest overall revenue
• Beverage products dominate customer purchases
• Bakery items contribute to cross-selling opportunities

Business Impact

Supports product optimization and menu development strategies.

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3. Weekday Sales Analysis

Sales trends were analyzed across weekdays to understand customer behavior.

Key Findings

• Customer traffic differs throughout the week
• Certain weekdays experience higher sales volumes
• Weekend purchasing behavior varies from weekdays

Business Impact

Improves workforce scheduling and promotional campaign planning.

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4. Top Product Analysis

Top-performing products were identified based on revenue and quantity sold.

Key Findings

- Espresso-based beverages performed strongly
- Premium coffee products generated high sales
- Some products consistently outperformed others

Business Impact

Supports inventory planning and pricing optimization.

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5. Store Performance Analysis

Revenue performance was evaluated across multiple store locations.

Key Findings

- Some branches consistently outperform others
- Store location significantly impacts revenue generation
- Customer traffic varies between locations

Business Impact

Helps management identify high-performing locations and operational improvement opportunities.

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6. Hourly Sales Analysis

Sales were analyzed by hour to understand peak operational periods.

Key Findings

- Morning hours generate the highest sales
- Sales peak during breakfast and commuting hours
- Afternoon sales gradually decline

Business Impact

Improves staffing management and operational efficiency.

Data Visualization Techniques Used

The dashboard includes several visualization methods commonly used in business intelligence reporting.

Visualization	Purpose
KPI Cards	Display business metrics

Visualization	Purpose
Line Charts	Analyze sales trends
Bar Charts	Compare products and stores
Pie Charts	Visualize category contribution
Pivot Tables	Summarize data
Slicers	Enable interactive filtering

Tools & Technologies Used

Tool	Purpose
Microsoft Excel	Data cleaning and dashboard development
Pivot Tables	Data aggregation and analysis
Pivot Charts	Visualization and reporting
Excel Formulas	KPI calculations
Business Intelligence Techniques	Insight generation

Business Insights Generated

The project produced several meaningful business insights:

- Coffee products contribute the highest percentage of revenue
- Morning hours are the busiest operational period
- Certain store branches significantly outperform others
- Customer purchasing patterns vary across weekdays
- Product category performance can support future business strategies
- KPI dashboards improve executive-level decision-making

Challenges Faced During the Project

Several challenges were encountered while developing the project:

- Cleaning large transactional datasets
- Managing inconsistent date and time formats
- Handling missing values and duplicates
- Creating meaningful KPIs
- Designing professional dashboards
- Managing large pivot table calculations

These challenges helped strengthen problem-solving and analytical skills.

Skills Demonstrated

Technical Skills

- Data Cleaning
- Excel Dashboard Development
- Pivot Tables
- Pivot Charts
- KPI Reporting
- Data Visualization
- Business Intelligence
- Sales Analytics

Analytical Skills

- Trend Analysis
- Business Reporting
- Problem Solving
- Insight Generation
- Performance Analysis

Future Improvements

The project can be further enhanced by:

- Integrating Power BI dashboards
- Building sales forecasting models
- Performing customer segmentation analysis
- Implementing predictive analytics
- Automating reporting processes
- Adding profitability and margin analysis
- Using SQL for database-level cleaning
- Developing Python-based analytical workflows

Conclusion

The Coffee Shop Sales Analysis & Business Intelligence Dashboard project successfully transformed raw transactional sales data into meaningful and actionable business insights.

Using data cleaning, KPI reporting, dashboard visualization, and business analytics techniques, the project provides a comprehensive overview of operational and financial performance.

The dashboard supports data-driven decision-making by helping management analyze revenue trends, customer purchasing behavior, product performance, and branch-level operations.

This project demonstrates practical business intelligence and analytics skills aligned with real-world industry practices and showcases the ability to develop professional-level reporting solutions.

References

- Microsoft Excel Documentation
- Business Intelligence Dashboard Practices
- Sales Analytics Reporting Techniques
- Data Visualization Best Practices

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